# Topic 25 — Global Environmental Geography

**Covers syllabus (click to expand)**

Global Environmental Geography (biomes · climate zones · world deserts · floral regions · hotspots)

**Sources baked in:** NCERT Geo 11; Bio 12 Ch 15–16; Köppen; Myers hotspots; UPPCS PYQs 2018–2025
**Weight:** ★★ Medium–High — biome matching; desert countries; Brazil Selva; tribe–habitat traps
**Last verified:** August 2026

## Current Affairs (this topic)

- Biome (जीवोम) and realm stems stay geographic; CA rarely moves the Köppen (कोपेन) or biome map.
- Bridge biome teaching with Ecology (पारिस्थितिकी) & Ecosystem (पारिस्थितिकी तंत्र) without duplicating full cards.

## 25.1 Scope & Climate Controls

**Identity:** Spatial environment (पर्यावरण) patterns | Climate primary control | Latitude + continentality + altitude

- **Global environmental geography** studies *where* environmental systems occur and *why* — linking physical geography (climate, relief, oceans) to biomes, biodiversity, and degradation.
- **Climate** (temperature + precipitation) is the primary control on biome distribution.
- **Latitude:** equatorial regions receive direct sunlight year-round → high productivity and biodiversity; polar regions receive oblique rays → low productivity.
- **Continentality:** interior continents (central Australia, central Asia) have extreme temperatures and aridity.
- **Altitude:** Himalaya (हिमालय), Andes, Alps show vertical zonation — tropical forest at base, alpine/nival at summit.
- **Ocean currents:** warm currents (Gulf Stream) mild climate; cold currents (Humboldt) support fisheries and coastal deserts (Atacama).
- **Subtropical highs (horse latitudes (हॉर्स अक्षांश) ~20–35° N/S)** suppress rainfall — major desert belt (Sahara, Arabian, Thar, Australian interior).
- **Latitudinal biodiversity gradient:** species richness highest near equator, decreases toward poles.

**PYQ — UPPCS Prelims 2025, Q1**

Assertion (A): The interior part of Australia is desert and semi-desert. Reason (R): Northern Australia is situated in the temperate zone.

Options: A. Both true; R does not explain A B. A false, R true C. A true, R false D. Both true; R explains A

**Show answer**

**Ans: C** — Interior aridity from **subtropical high + continentality**. Northern Australia is largely **tropical**, not temperate.

**Logic:** Australia desert ≠ temperate-zone explanation.

## 25.2 Major Global Biomes

**Identity:** Climate-defined regions | Rainforest → desert → grassland → taiga → tundra | Mangrove transitional

- A **biome** is a large region with characteristic vegetation and fauna shaped by climate.
- **Tropical rainforest:** hot, wet year-round; layered canopy; highest biodiversity — Amazon (**Selva**), Congo, SE Asia, Western Ghats (पश्चिमी घाट).
- **Tropical deciduous:** wet–dry monsoon — sal/teak forests of peninsular India.
- **Desert:** typically **<25 cm** rain; sparse xerophytes — Sahara, Arabian, Gobi, Thar, Atacama.
- **Savanna:** tropical grassland with **scattered trees** and distinct wet/dry seasons — East Africa ("Land of Big Games"); Gir (गीर) (Gujarat) is India patch.
- **Temperate grassland:** few trees; fertile soils — Prairies (N America), Steppes (Eurasia), **Pampas (Argentina)**, **Puszta (Hungary)**.
- **Taiga (boreal):** cold winters; spruce/fir/pine; **permafrost patches** — Siberia, Canada, Scandinavia.
- **Tundra:** short summer; moss/lichen; **continuous permafrost** — Arctic coasts; no trees.
- **Mediterranean (chaparral):** wet winters, dry summers; fire-adapted shrubs — California, Mediterranean Basin, Cape Town.
- **Mangrove / wetland:transitional** terrestrial–aquatic — not purely terrestrial.

| Biome | Climate | Global example |
|---|---|---|
| Tropical rainforest | Hot wet year-round | Amazon Selva |
| Desert | Arid <25 cm | Sahara, Gobi, Thar |
| Savanna | Wet–dry grassland + trees | East Africa |
| Temperate grassland | Moderate (नरम दल) rain | Pampas, Puszta |
| Taiga | Cold coniferous | Siberia, Canada |
| Tundra | Very cold, permafrost | Arctic |

**Logic:** Permafrost = Taiga + Tundra**, not Savannah.** 14 global biomes ≠ 10 Indian biogeographic zones**.

## 25.3 Climate Zones (Köppen)

**Identity:** A–B–C–D–E | India Am | Horse latitudes = deserts | ITCZ drives monsoon

- **Köppen Group A (Tropical):** all months ≥18°C — Af rainforest, **Am monsoon**, Aw savanna. India mainly **Am**.
- **Group B (Arid):** evaporation > precipitation — BW desert, BS steppe. Thar **BWh** (hot); Gobi **BWk** (cold).
- **Group C (Temperate):** mild winters — Mediterranean Csa, humid subtropical Cwa.
- **Group D (Cold/Continental):** cold winters — taiga belt Siberia/Canada.
- **Group E (Polar):** always cold — tundra ET, ice cap EF (Antarctica).
- **ITCZ** seasonal shift drives monsoon and equatorial rain; failure causes Sahel drought (सूखा).
- Same latitude can differ: cold west-coast currents → aridity; warm east coasts → wetter.

| Group | Name | India / example |
|---|---|---|
| A | Tropical | India Am |
| B | Arid | Thar BWh, Gobi BWk |
| C | Temperate | Mediterranean climates |
| D | Cold | Siberia taiga |
| E | Polar | Antarctica |

> **Logic:** "India has Af rainforest climate everywhere" = **FALSE** — India is mainly **Am**.

## 25.4 World Deserts & Critical Ecosystems

**Identity:** Sahara largest hot | Atacama driest | Gobi Mongolia+China | Brazil Selva + Terra Roxa

- **Sahara (Africa):** largest **hot desert**.
- **Atacama (Chile):** among the **driest** deserts on Earth — cold Humboldt Current + rain shadow (छाया).
- **Gobi:** cold desert extending mainly into **Mongolia and China**.
- **Kalahari:** southern Africa — **Bushman (San)** habitat.
- **Thar:** India–Pakistan hot desert — subtropical high + rain shadow.
- **Australian interior:** desert/semi-desert under subtropical high and continentality.
- **Amazon (Brazil):** equatorial rainforest called **Selva** — among Earth's richest biodiversity zones.
- **Terra Roxa:** fertile dark volcanic (basalt) soil of Brazil — coffee/soy region (often tested with Selva).

**PYQ — UPPCS Prelims 2025, Q27**

The Gobi Desert extends into: (1) Mongolia (2) China (3) Russia (4) Kazakhstan

Options: A. Only 1 and 2 B. Only 1, 2 and 4 C. Only 1, 2 and 3 D. Only 1

**Show answer**

**Ans: A** — Primary extent = **Mongolia + China**.

**PYQ — UPPCS Prelims 2025, Q63**

Brazil: (1) Equatorial rainforests called "Selva". (2) Fertile dark soil known as "Terra Roxa".

Options: A. Only 2 B. Neither C. Both D. Only 1

**Show answer**

**Ans: C** — Both statements correct.

**PYQ — UPPCS Prelims 2025, Q23**

Tribe–habitat NOT correctly matched: Bushman–Kalahari; Eskimo–N Canada; Pygmy–Congo; Masai–West Africa

**Show answer**

**Ans: B (Only 4)** — Masai live in **East Africa** (Kenya/Tanzania), not West Africa.

| Desert | Region | Fact |
|---|---|---|
| Sahara | N Africa | Largest hot desert |
| Atacama | Chile | Driest |
| Gobi | Mongolia + China | Cold desert |
| Kalahari | S Africa | Bushman/San |
| Thar | India–Pakistan | Hot desert |

> **Logic:** Pampas = **Argentina** (2018). Land of Big Games = **Tropical Savanna**.

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## 25.5 Floral Regions & Vegetation Matching

**Identity:** Engler floral regions | Sauer planting hearth | Puszta Hungary | Vegetation–region pairs

- **World Floral Regions** classification is associated with **Adolf Engler**.
- **Planting Hearth** (agricultural origin centres) is associated with **C.O. Sauer**.
- **Cryophyte Steppe** = cold-region steppe vegetation of the **Arctic Tundra** belt.
- **Puszta Grassland** = temperate grassland of **Hungary**.
- Classic vegetation–region pairs: **Epiphytes–Equatorial**; **Acacia–Savanna**; **Baobab–Savanna**; **Cedars–Mediterranean**.

**PYQ — UPPCS Prelims 2024, Q29**

Match: A. Planting Hearth B. Cryophyte Steppe C. World Floral Regions D. Puszta Grassland With: 1. Hungary 2. Adolf Engler 3. C.O. Sauer 4. Arctic Tundra

Options: A. 3 4 2 1 B. 2 1 4 3 C. 3 4 1 2 D. 2 3 4 1

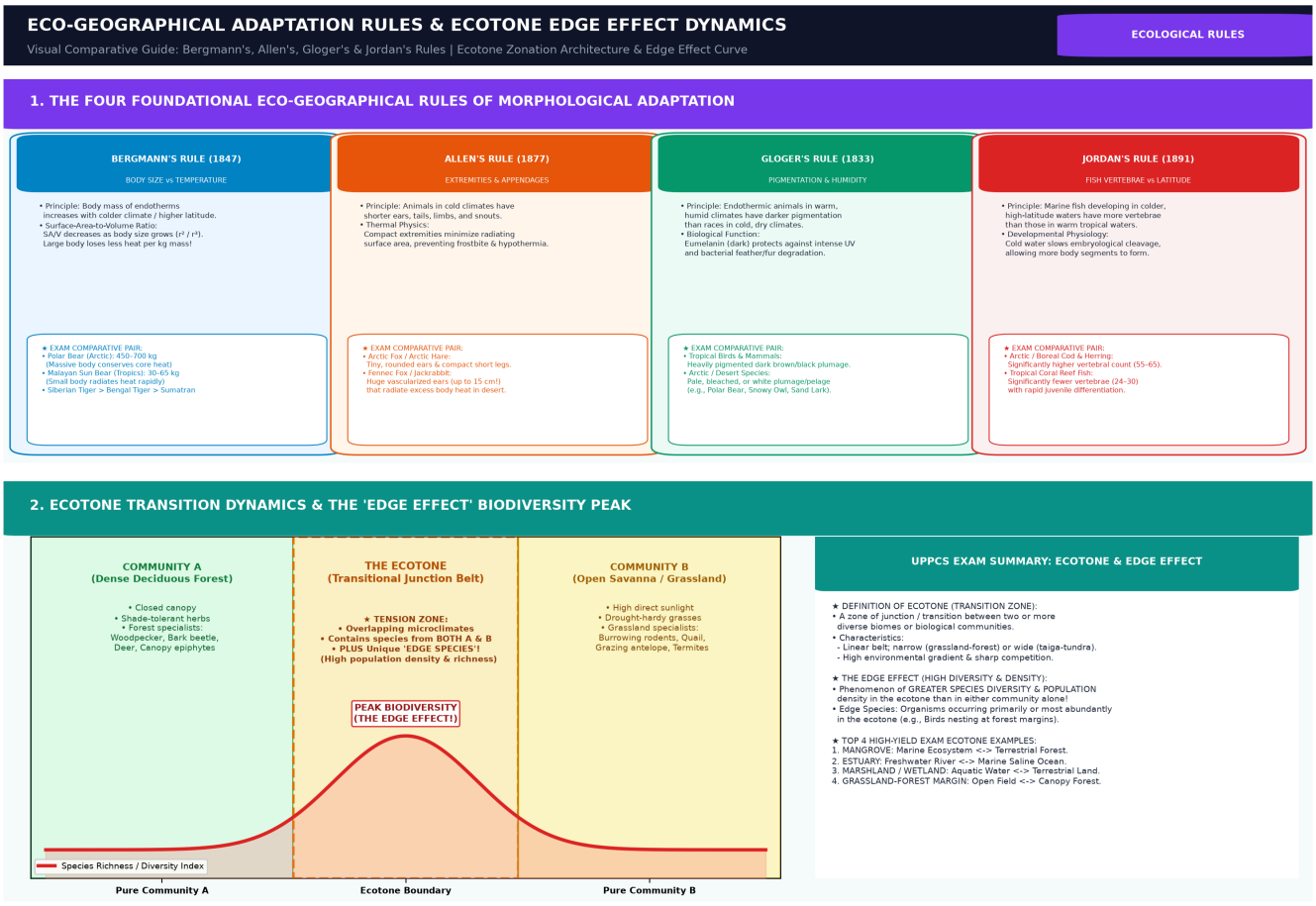
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**Ans: A** — Sauer (3), Arctic Tundra (4), Engler (2), Hungary (1).

| Pair | Fact |
|---|---|
| Planting Hearth | C.O. Sauer |
| World Floral Regions | Adolf Engler |
| Cryophyte Steppe | Arctic Tundra |
| Puszta | Hungary |
| Pampas | Argentina |

> **Logic:** Do not swap Engler ↔ Sauer.

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### Eco-Geographical Rules & Ecotone Dynamics

- **Bergmann's rule:** Body size in endotherms increases with colder climate (higher latitude) to reduce heat loss per unit volume (e.g., Polar Bear > Sun Bear).
- **Allen's rule:** Extremities and appendages (ears, limbs, snouts) are shorter and more compact in colder climates (e.g., Arctic Fox short ears vs Desert Fennec Fox large ears).
- **Gloger's rule:** Endotherms in warm, humid climates develop darker melanin pigmentation (eumelanin) than those in cold, arid regions.
- **Jordan's rule:** Marine fish developing in colder, high-latitude waters have more vertebrae than relatives in warm tropical waters.
- **Ecotone (संक्रमिका):** A transition zone where two biomes overlap (e.g., Mangrove between sea and land; Estuary between river and ocean).
- **Edge effect (किनारा प्रभाव):** Greater species diversity and population density occur in the ecotone than in either adjacent community alone, harboring unique "edge species".

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## 25.6 Biodiversity Hotspots & Degradation Belts

**Identity:** Myers criteria | 36 hotspots | 4 in India | Pollution corridors

- **Biodiversity hotspot (जैव विविधता हॉटस्पॉट) (Myers):** high **endemism** + ≥**70%** natural habitat loss.
- About **36** global hotspots. **Four (चातुर्याम) in India:** Himalaya, Western Ghats–Sri Lanka, Indo-Burma (NE), Sundaland (Nicobar (निकोबार)).
- **Megadiversity countries:** 17 globally including India, Brazil, Indonesia, China, Australia.
- **Global degradation belts:** Indo-Gangetic air pollution corridor; Amazon deforestation arc; Niger Delta (डेल्टा) oil pollution; plastic ocean gyres (Great Pacific Garbage Patch).
- Environmental problems are often **transboundary** — Saharan dust to Europe, Amazon fires affecting global $CO_2$, plastic gyres crossing EEZs.

| India hotspot | Region |
|---|---|
| Himalaya | Mountain endemism |
| Western Ghats | Peninsular rainforest |
| Indo-Burma | NE India |
| Sundaland | Nicobar |

> **Logic:** Hotspot needs **both** endemism and habitat loss — not richness alone.

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